

Class (Section): _____

Name: _____

Roll No: _____

Date: _____

Natural Language Processing (NLP)

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ACTIVITY SHEET 1 BYTE PAIR ENCODING (BPE) & Word Piece

Training Text:

low low lower newest wide

1. Step 1 — Write Each Word as Characters +

Word	Split Into Characters
low	
low	
lower	
newest	
wide	

2. Step 2 — Count All Adjacent Pairs (Iteration 1)

Fill frequencies:

Pair	Count
(l,o)	
(o,w)	
(w,))	
(w,e)	
(e,r)	
(e,w)	
(n,e)	
(e,s)	
(s,t)	
(w,i)	
(i,d)	
(d,e)	
(e,.)	

3. Step 3 — Most Frequent Pair

Most frequent pair =

(_____ , _____)

4. Step 4 — MERGE #1

Merged token:

Updated corpus (write merged version for each word):

Word	After Merge #1
low	
low	
lower	

newest	
wide	

5. Step 5 — Iteration 2 Pair Counts

Pair	Count
...	...

6. MERGE #2

Merge pair:

(_____ , _____) → _____

Update corpus:

Word	After Merge #2
low	
low	
lower	
newest	
wide	

7. Continue for MERGE #3, #4, #5...

You write on board → they fill blanks.

8. Final BPE Vocabulary

Students fill:

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

WORDPIECE – Student Activity Sheet

Training Text:

low low lower newest wide

1. Step 1 — Write Each Word as Characters

Word	Characters
low	

low	
lower	
newest	
wide	

2. Step 2 — Character Frequencies (Students fill)

Character	Count
l	
o	
w	
e	
r	
n	
s	
t	
i	
d	

3. Step 3 — Pair Frequencies

Pair	Count
(l,o)	
(o,w)	
(w,e)	
(e,r)	
(n,e)	
(e,w)	
(s,t)	
(w,i)	
(i,d)	
(d,e)	

4. Step 4 — WordPiece Score Calculation

Formula:

$$\text{score} = \text{freq}(\text{pair}) / \text{freq}(\text{left_symbol})$$

Pair	freq(pair)	freq(left)	score	highest?
(l,o)				
(o,w)				
(w,e)				
(e,r)				
(d,e)				

5. MERGE #1

Merge:

(_____ , _____) → _____

Updated corpus:

Word	After Merge #1
low	
low	
lower	
newest	
wide	

6. MERGE #2 + MERGE #3 (students fill as you teach)

Provide space:

MERGE #2: (_____ , _____) → _____

Updated corpus: _____

MERGE #3: (_____ , _____) → _____

Updated corpus: _____

7. Final: WordPiece Tokenization Practice

Students fill tokenization using BERT rules:

- first token: no prefix
- continuation: prefix ##

Word	WordPiece Tokens (students fill)
low	
lower	
newest	
wide	
lowest	